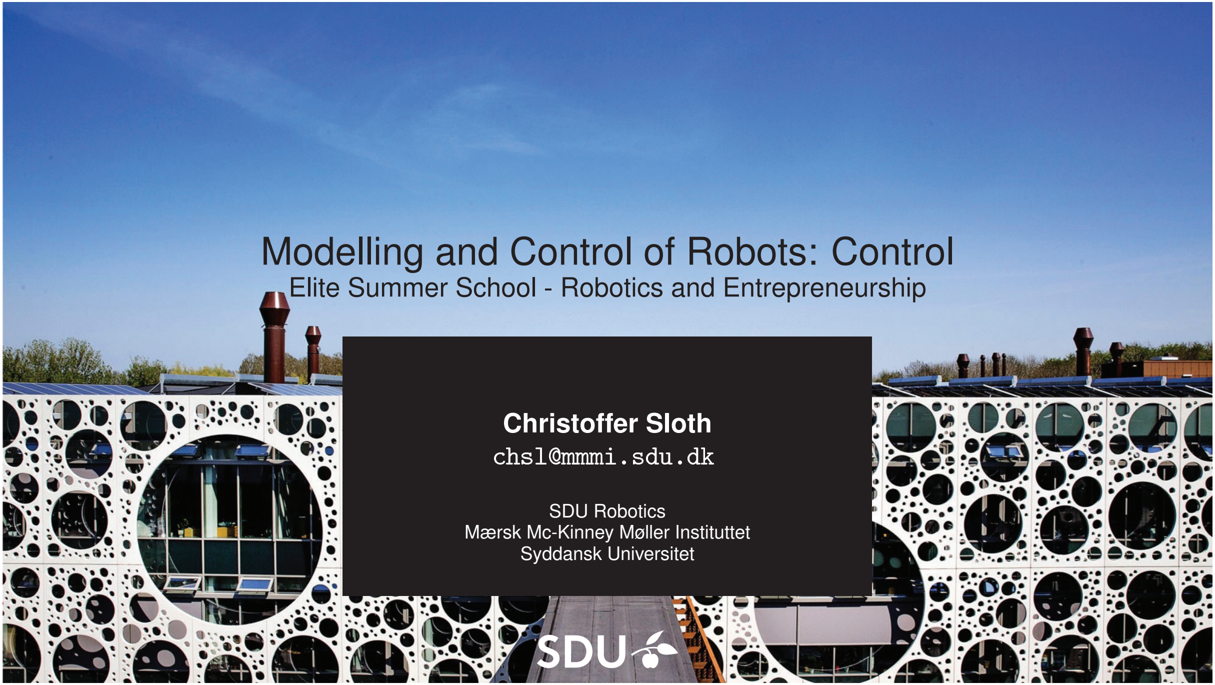

Modelling and Control of Robots: Control

Elite Summer School - Robotics and Entrepreneurship

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Agenda

Motion Control
- Introduction
- Inverse Dynamic Control in Joint Space
- Inverse Dynamic Control in Operational Space

Indirect Force Control
- Introduction
- Impedance Control
- Admittance Control

Kinesthetic Teaching
- Introduction
- Learning In-Contact Tasks

Summary

We address how to make a robot follow a desired motion. For this, we consider

- Joint space control
- Operational space control

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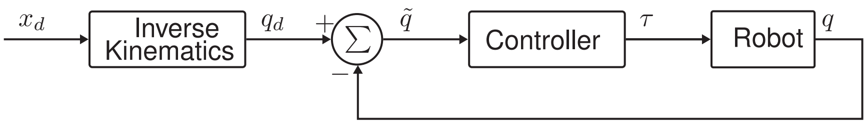
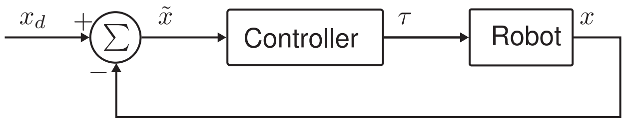
A desired motion is often specified in the operational space (this is called x_d), but actuators are controlled by applying generalized forces τ at the actuators.

For ***joint space control*** one has to specify the desired motion in joint space (as configurations). This is accomplished as follows

1. Compute the desired joint motion q_d from x_d
2. Control generalized forces τ to minimize joint space error $\tilde{q} = q_d - q$, where q is the actual joint trajectory



When considering ***operational space control***, the kinematics is embedded in the control and the desired operational space motion x_d is controlled directly.



It is favorable to control the robot in operational space when the robot pose x is measured directly; otherwise, the kinematic error is not minimized.

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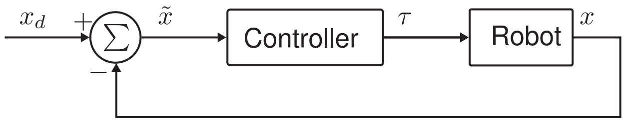

It is favorable to control the robot in operational space when the robot pose x is measured directly; otherwise, the kinematic error is not minimized.

Remark that the controller minimizes the error in operational space, i.e. $\tilde{x} = x_d - x$.

We consider robot models in both joint space and operational space.

A **joint space** dynamic model of an n degree of freedom manipulator is given by

$$B(q)\ddot{q} + \underbrace{C(q,\dot{q})\dot{q} + \tau_f(\dot{q}) + g(q)}_{=n(q,\dot{q})} = \tau - J^T(q)h_e$$

where q is the configuration, $B(q)$ is the mass matrix, $C(q,\dot{q})$ contain centrifugal and Coriolis terms, $g(q)$ contains torques caused by gravity, $\tau_f(\dot{q})$ is the friction torque, τ is the actuator torque, h_e is the external wrench.

An **operational space** dynamic model of an n degree of freedom manipulator is given by

$$\Lambda(q)\ddot{x} + \Gamma(q,\dot{q})\dot{x} + \eta(q) = h_c - h_e$$

where h_c is the commanded wrench in operational space and

$$\begin{aligned}\Lambda(q) &= J^{-T}(q)B(q)J^{-1}(q)\\ \Gamma(q,\dot{q}) &= J^{-T}(q)C(q,\dot{q})J^{-1}(q) - \Lambda(q)\dot{J}(q)J^{-1}(q)\\ \eta(q) &= J^{-T}(q)g(q)\end{aligned}$$

where $J(q)$ is the geometric Jacobian at configuration q.

The transfer function of a second-order system is

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

and is described by two parameters: $\zeta > 0$ and $\omega_n > 0$.

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The system has two poles, which are $s \in \mathbb{C}$ where

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The poles of

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

are values of s given by

$$s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4 \cdot 1 \cdot \omega_n^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.$$

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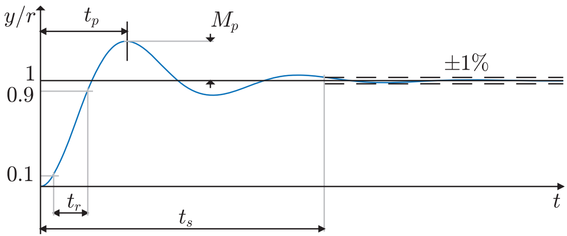
Terminology

- The parameter ζ is called the **damping ratio**.
- The parameter ω_n is called the **undamped natural frequency**.

We consider three different performance measures of dynamical systems

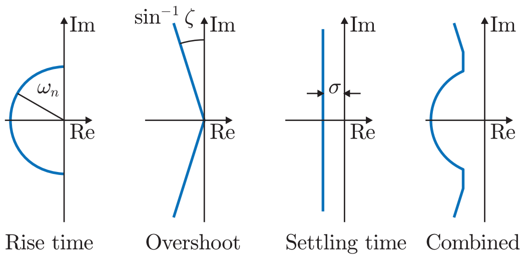
1. The **rise time** t_r.
2. The **settling time** t_s.
3. The **overshoot** M_p.



The **peak time** is denoted t_p.

Performance Specification

Frequency and Time Specifications



To obtain a rise rise shorter than t_r

$$\omega_n \geq \frac{1.8}{t_r}$$

To obtain an overshoot that is smaller than M_p

$$\zeta \geq \sqrt{\frac{(\frac{\log(M_p)}{-\pi})^2}{1+(\frac{\log(M_p)}{-\pi})^2}}$$

To obtain an α%-settling time shorter than t_s

$$\sigma \geq \frac{-\log(\alpha/100)}{t_s}$$

Problem 1
Given a joint space dynamic model of an n degree of freedom manipulator

$$B(q)\ddot{q} + \underbrace{C(q,\dot{q})\dot{q} + F_v\dot{q} + g(q)}_{=n(q,\dot{q})} = \tau$$

Find a control law that transforms the system model into a double integrator system

$$\ddot{q} = y.$$

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Problem 2
Given a double integrator system

$$\ddot{q} = y$$

where $y \in \mathbb{R}^n$ is the controllable input. Find a control law such that q tracks a desired trajectory q_{d}.

The following control law is known as an ***inverse dynamic controller***

$$\tau = B(q)y + C(q, \dot{q})\dot{q} + F_v\dot{q} + g(q)$$

where y is a controllable input solves Problem 1.

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$$B(q)(\ddot{q} - y) = 0.$$

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By applying the inverse dynamic controller to system, we obtain the following relation

$$B(q)(\ddot{q} - y) = 0.$$

Since the mass matrix $B(q)$ is nonsingular, we conclude that

$$\ddot{q} = y.$$

Inverse Dynamic Control

Solution to Problem 2

Given any desired trajectory $q_\mathsf{d}(t)$. The control law

$$y = \ddot{q}_\mathsf{d} + K_\mathsf{D}(\dot{q}_\mathsf{d} - \dot{q}) + K_\mathsf{P}(q_\mathsf{d} - q),$$

where K_D and K_P are positive definite matrices, solves Problem 2.

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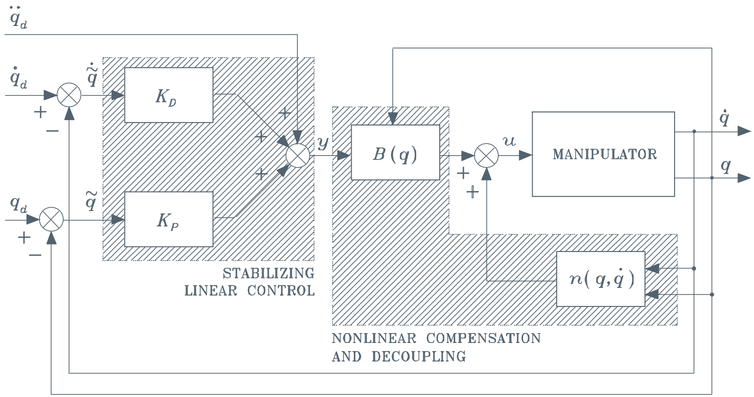
By applying the control law to the double integrator system, we obtain

$$\ddot{q} = \ddot{q}_\mathsf{d} + K_\mathsf{D}(\dot{q}_\mathsf{d} - \dot{q}) + K_\mathsf{P}(q_\mathsf{d} - q),$$

which can be rewritten as

$$\ddot{\tilde{q}} + K_\mathsf{D}\dot{\tilde{q}} + K_\mathsf{P}\tilde{q} = 0,$$

where $\tilde{q} = q_\mathsf{d} - q$ is the tracking error. It is known that the solution $\tilde{q}$ converges to zero (it is a stable linear system when all gains are positive); hence, the tracking error converges to zero.



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$\dot{q}_d$
$\dot{\tilde{q}}$
K_D
q_d
$\tilde{q}$
K_P
y
$B(q)$
u
MANIPULATOR
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LINEAR CONTROL
NONLINEAR COMPENSATION
AND DECOUPLING

The selection of K_P and K_D can be accomplished from the error dynamics

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By Laplace transform, we obtain the 2nd order transfer function

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Therefore, $\omega_n^2 = K_P$ and $2\zeta\omega_n = K_D$ and hence

$$t_r \approx \frac{1.8}{\sqrt{K_P}}$$

$$t_s \approx \frac{-\log(\alpha/100)}{K_D/2}$$

$$\zeta = \frac{K_D}{2\sqrt{K_P}}$$

Problem 3
Given a operational space dynamic model of an n degree of freedom manipulator

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where $\alpha \in \mathbb{R}^n$ is the controllable input. Find a control law such that x tracks a desired trajectory x_d.

Inverse Dynamic Control
Solution to Problem 3

The following control law is known as an ***inverse dynamic controller*** in operational space

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Since the pseudo inertia tensor $\Lambda(q)$ is nonsingular, we conclude that

$$\ddot{x} = \alpha.$$

Force control strategies can be classified as

- Indirect force control (control not directly having a force feedback loop)
  - Compliance control
  - Impedance control
  - Admittance control
- Direct force control
  - Force control
  - Hybrid force/motion control

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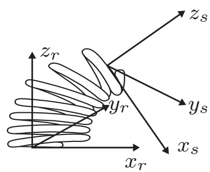
Force control usually works in operational space - it is an operational space control.

Let $dx_{r,s}$ be the displacement from the equilibrium of frame s

$$dx_{r,s} = \begin{bmatrix} do_{r,s} \\ \omega_{r,s} dt \end{bmatrix} = v_{r,s} dt$$



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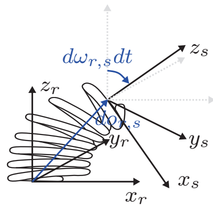

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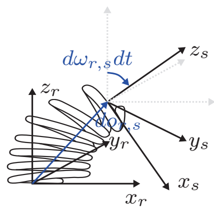

The elastic force (wrench) of a spring h_s between the two bodies is

$$h_s = K dx_{r,s}$$

where $K = K^T$ is a positive semidefinite stiffness matrix.

The generalization of Hook's law is

$$h_s = \begin{bmatrix} f_s \\ \mu_s \end{bmatrix} = \underbrace{\begin{bmatrix} K_f & K_c \\ K_c^T & K_\mu \end{bmatrix}}_{=K} \begin{bmatrix} do_{r,s} \\ \omega_{r,s} dt \end{bmatrix}$$

where K_f is the translational stiffness [N/m], K_μ is the rotational stiffness [Nm/rad], and K_c is the coupling stiffness [N/rad].

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where K_f is the translational stiffness [N/m], K_μ is the rotational stiffness [Nm/rad], and K_c is the coupling stiffness [N/rad].

The compliance is the reciprocal of the stiffness, i.e.,

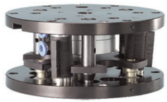
$$dx_{r,s} = Ch_s$$

where C is the ***compliance matrix***.

The compliance can origin from mechanical components (**passive compliance**) or by software components (**active compliance**).

In this course only active compliance is considered by means of robot control.

A ***remote center of compliance*** (RCC) is a mechanical device that makes the coupling stiffness K_c symmetric or null.



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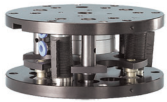
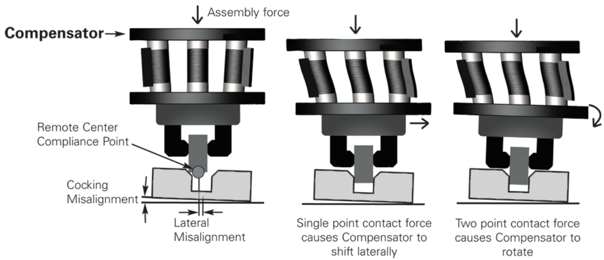
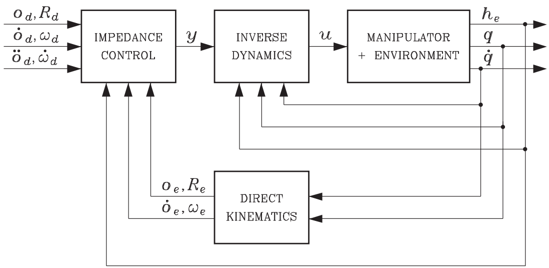


This device is used for assembly, where the remote center of compliance point is put at the end of the manipulated part.



An impedance control is an indirect force control that is designed to comply with the environment. Impedance control referes to a mechanical impedance, which is given by a mass-spring-damper system.

A block diagram of an impedance control is given in the following.



An impedance control can be implemented without measuring the contact force as

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) \\ y = J_A^{-1}(q)M_d^{-1}\left(M_d\ddot{x}_d + K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_A(q,\dot{q})\dot{q}\right) \end{cases}$$

where M_d is a desired mass matrix.

Impedance Control
Without Force Measurement

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where M_d is a desired mass matrix. The closed-loop dynamics are then

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = M_dB_A^{-1}(q)h_A$$

where

$$B_A(q) = J_A^{-T}(q)B(q)J_A^{-1}(q)$$

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where

$$B_A(q) = J_A^{-T}(q) B(q) J_A^{-1}(q)$$

Due to the configuration-dependent matrix $B_A(q)$, the interaction is in general nonlinear and coupled. The mechanical properties of the robot determine how the coupling is.

If linearity and decoupling is needed, the it is necessary to measure the contact force.

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If the force h_e is measured then the control can be modified as

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Then the closed-loop dynamics are

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = h_A$$

where $h_A = T_A^T(x_e)h_e$.

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where $h_A = T_A^T(x_e)h_e$.

The force h_A has a configuration-dependent relation to h_e; thus, the impedance depends on the current end-effector orientation.

Under the assumption that the desired frame O_d is time-varying

$$\dot{\tilde{x}} = -J_{A_d}(q,\tilde{x})\dot{q} + \underbrace{\begin{bmatrix} R_d^T \dot{o}_d + S(\omega_d^d) o_{d,e}^d \\ T^{-1}(\phi_{d,e})\omega_d^d \end{bmatrix}}_{= b(\tilde{x}, R_d, \dot{o}_d, \omega_d)}$$

Impedance Control
Time-Varying Reference Frame

Under the assumption that the desired frame O_d is time-varying

$$\dot{\tilde{x}} = -J_{A_d}(q,\tilde{x})\dot{q} + \underbrace{\begin{bmatrix} R_d^T \dot{o}_d + S(\omega_d^d) o_{d,e}^d \\ T^{-1}(\phi_{d,e})\omega_d^d \end{bmatrix}}_{=b(\tilde{x},R_d,\dot{o}_d,\omega_d)}$$

Then

$$\ddot{\tilde{x}} = -J_{A_d}(q,\tilde{x})\ddot{q} - \dot{J}_{A_d}(q,\tilde{x})\dot{q} + \dot{b}$$

Under the assumption that the desired frame O_d is time-varying

$$\dot{\tilde{x}} = -J_{A_d}(q,\tilde{x})\dot{q} + \underbrace{\begin{bmatrix} R_d^T\dot{o}_d + S(\omega_d^d)o_{d,e}^d \\ T^{-1}(\phi_{d,e})\omega_d^d \end{bmatrix}}_{=b(\tilde{x},R_d,\dot{o}_d,\omega_d)}$$

Then

$$\ddot{\tilde{x}} = -J_{A_d}(q,\tilde{x})\ddot{q} - \dot{J}_{A_d}(q,\tilde{x})\dot{q} + \dot{b}$$

A new control law

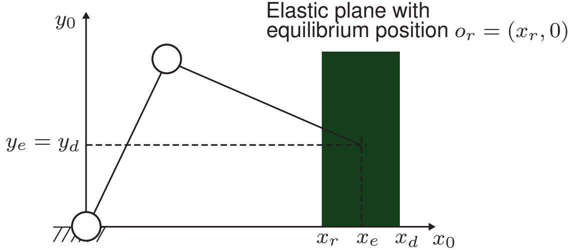
$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) + J^T(q)h_e \\ y = J_{A_d}^{-1}M_d^{-1}\left(K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_{A_d}\dot{q} + M_d\dot{b} - h_e^d\right) \end{cases}$$

gives closed-loop dynamics

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = h_e^d$$

where all the vectors are referred to the desired frame.

Consider again the system shown below (only translation).



The controlled system (impedance control in contact with compliant environment) has dynamics

$$\underbrace{\begin{bmatrix} m_{dx} & 0 \\ 0 & m_{dy} \end{bmatrix}}_{M_d} \begin{bmatrix} \ddot{x}_e \\ \ddot{y}_e \end{bmatrix} + \underbrace{\begin{bmatrix} k_{Dx} & 0 \\ 0 & k_{Dy} \end{bmatrix}}_{K_D} \begin{bmatrix} \dot{x}_e \\ \dot{y}_e \end{bmatrix} + \underbrace{\begin{bmatrix} k_{Px} & 0 \\ 0 & k_{Py} \end{bmatrix}}_{K_P} \begin{bmatrix} x_e - x_d \\ y_e - y_d \end{bmatrix} = \underbrace{\begin{bmatrix} k_x & 0 \\ 0 & 0 \end{bmatrix}}_{K} \begin{bmatrix} x_r - x_e \\ y_r - y_e \end{bmatrix}$$

The controlled system (impedance control in contact with compliant environment) has dynamics

$$\underbrace{\begin{bmatrix} m_{dx} & 0 \\ 0 & m_{dy} \end{bmatrix}}_{M_d} \begin{bmatrix} \ddot{x}_e \\ \ddot{y}_e \end{bmatrix} + \underbrace{\begin{bmatrix} k_{Dx} & 0 \\ 0 & k_{Dy} \end{bmatrix}}_{K_D} \begin{bmatrix} \dot{x}_e \\ \dot{y}_e \end{bmatrix} + \underbrace{\begin{bmatrix} k_{Px} & 0 \\ 0 & k_{Py} \end{bmatrix}}_{K_P} \begin{bmatrix} x_e - x_d \\ y_e - y_d \end{bmatrix} = \underbrace{\begin{bmatrix} k_x & 0 \\ 0 & 0 \end{bmatrix}}_{K} \begin{bmatrix} x_r - x_e \\ y_r - y_e \end{bmatrix}$$

The dynamics are given by second order systems. Hence, it is characterized by the a natural frequency ω_n and a damping ratio ζ.

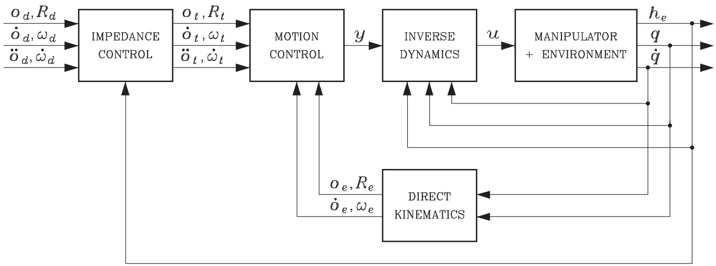
x-direction

$$\omega_{nx} = \sqrt{\frac{k_{P_x} + k_x}{m_{dx}}} \qquad \zeta_x = \frac{k_{D_x}}{2\sqrt{m_{dx}(k_{Px} + k_x)}}$$

y-direction

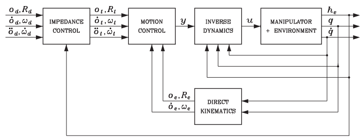
$$\omega_{ny} = \sqrt{\frac{k_{P_y}}{m_{dy}}} \qquad \zeta_y = \frac{k_{D_y}}{2\sqrt{m_{dy} k_{Py}}}$$

An admittance control is illustrated in the following figure.



Admittance Control
Definition

An admittance control is illustrated in the following figure.



The block *Impedance Control* is given by

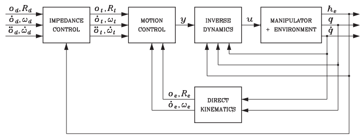
$$M_t \ddot{\tilde{z}} + K_{Dt} \dot{\tilde{z}} + K_{Pt} \tilde{z} = h_e^d$$

where M_t, K_{Dt}, K_{Pt} are the parameters of a mechanical impedance and $\tilde{z}$ is the operational space error, which is the output of *Impedance Control* and is given by

$$\tilde{z} = - \begin{bmatrix} o_{d,t}^d \\ \phi_{d,t} \end{bmatrix}$$

where $R_t^d = R_d^T R_t$ and $o_{d,t}^d = R_d^T(o_t - o_d)$.

An admittance control is illustrated in the following figure.



Motion Control and *Inverse Dynamics* are given by

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) + J^T(q)h_e \\ y = J_A^{-1} M_d^{-1} \left(M_d \ddot{x}_t + K_D \dot{\tilde{x}} + K_P \tilde{x} - M_d \dot{J}_A \dot{q} \right) \end{cases}$$

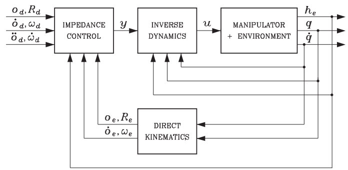
where

$$\tilde{x} = -\begin{bmatrix} o^t_{t,e} \\ \phi_{t,e} \end{bmatrix}$$

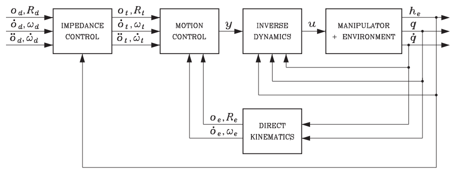
Why are both impedance and admittance controllers considered?

- Only a few industrial robots allow joint torque control; hence, only admittance control can be implemented.
- The impedance control is sensitive to model errors.
- The admittance control may become unstable when used for interacting with stiff environments.

Impedance



Admittance



Consider that the system model is affected by an additive disturbance τ_d, i.e.

$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + \tau_f(\dot{q}) + g(q) = \tau - J^T(q)h_e + \tau_d.$$

Consider that the system model is affected by an additive disturbance τ_d, i.e.

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + \tau_f(\dot{q}) + g(q) = \tau - J^T(q)h_e + \tau_d.$$

Then the dynamics of the **impedance control** is given by (in s-domain)

$$x_e(s) = x_d(s) - \frac{1}{M_d s^2 + K_D s + K_P}(h_e + h_d)$$

Consider that the system model is affected by an additive disturbance τ_d, i.e.

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + \tau_f(\dot{q}) + g(q) = \tau - J^T(q)h_e + \tau_d.$$

Then the dynamics of the **impedance control** is given by (in s-domain)

$$x_e(s) = x_d(s) - \frac{1}{M_d s^2 + K_D s + K_P}(h_e + h_d)$$

and the dynamics of the **admittance control** is given by (in s-domain)

$$x_e(s) = x_d(s) - \frac{1}{M_t s^2 + K_{D_t} s + K_{P_t}} h_e - \frac{1}{M_d s^2 + K_D s + K_P} h_d$$

where

$$h_d = -M_d J_A(q) B^{-1}(q) \tau_d.$$

Consider that the system model is affected by an additive disturbance τ_d, i.e.

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + \tau_f(\dot{q}) + g(q) = \tau - J^T(q)h_e + \tau_d.$$

Then the dynamics of the **impedance control** is given by (in s-domain)

$$x_e(s) = x_d(s) - \frac{1}{M_d s^2 + K_D s + K_P}(h_e + h_d)$$

and the dynamics of the **admittance control** is given by (in s-domain)

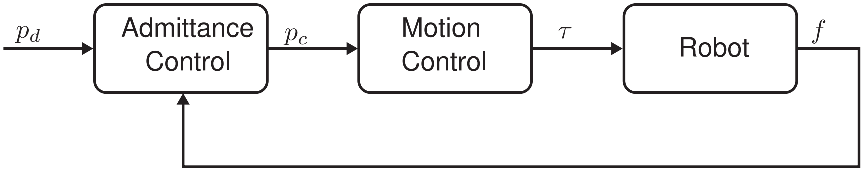
$$x_e(s) = x_d(s) - \frac{1}{M_t s^2 + K_{D_t} s + K_{P_t}} h_e - \frac{1}{M_d s^2 + K_D s + K_P} h_d$$

where

$$h_d = -M_d J_A(q) B^{-1}(q) \tau_d.$$

It is seen that the disturbance can be attenuated independently of the chosen compliance M_d, K_D, K_P only of the admittance control.

A simplified block diagram of an admittance controller is shown below.



We use this to define an admittance control with orientation represented using quaternion $q = (\eta, \epsilon)$, where η is the scalar part and ϵ in the vector part.

The translational part of the admittance controller is given by

$$M_p \Delta \ddot{p}_{cd} + D_p \Delta \dot{p}_{cd} + K_p \Delta p_{cd} = f$$

where M_p, D_p, K_p are 3×3 matrices, $f \in \mathbb{R}^3$ is force given in Base frame, and $\Delta p_{cd} = p_c - p_d$.

The rotational part of the admittance controller is given by

$$M_o \Delta \dot{\omega}_{cd}^d + D_o \Delta \omega_{cd}^d + K_o' \epsilon_{cd}^d = \mu^d$$

where M_o, D_o, K_o are 3×3 matrices, $\mu^d \in \mathbb{R}^3$ is the torque applied to the end-effector given in desired frame, $\Delta \omega_{cd} = \omega_c - \omega_d$, and $\epsilon_{cd}^d = \eta_d \epsilon_c - \eta_c \epsilon_d - S(\epsilon_c)\epsilon_d$. The rotational stiffness matrix is

$$K_o' = 2E^T(\eta_{cd}, \epsilon_{cd}^d) K_o$$

where K_o is the stiffness in Euler angle representation and

$$E(\eta, \epsilon) = \eta I - S(\epsilon)$$

The angular velocity must be integrated into a quaternion $q = (\eta_{cd}^d, \epsilon_{cd}^d)$

$$q(t + dt) = \exp\left(\frac{dt}{2}\omega_{cd}^d(t + dt)\right) * q(t)$$

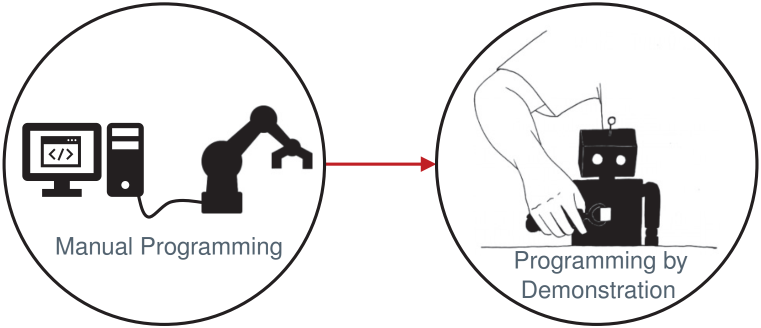
where

$$\exp(r) = (\eta, \epsilon) = (\cos(||r||), \frac{r}{||r||}\sin(r))$$

Kinesthetic Teaching

Programming by Demonstration

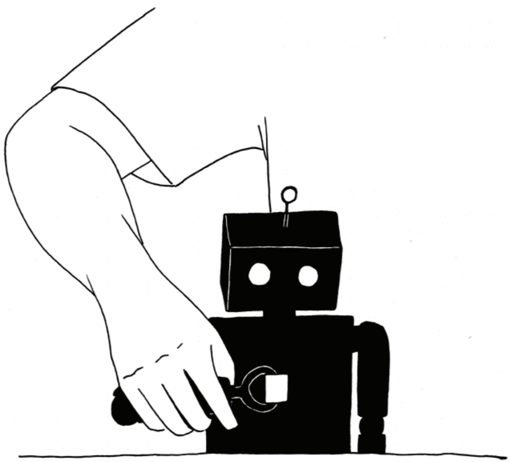
Programming by demonstration (PbD) is a no-code approach to programming robots that can be accomplished by people that do not have traditional programming skills.



Kinesthetic Teaching
Motivation

Program by demonstration is accomplished for the following reasons

- Makes teaching of **complex paths simple**.
- Enables **non-experts** to quickly and intuitively program robots.
- Makes it **simpler** to program certain kinds of tasks (e.g. assembly).
- Simplifies **parameter configuration**.
- Can **reduce changeover time** in production.



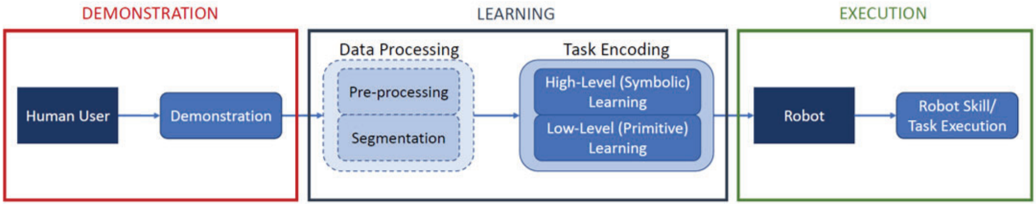
- \+ No mappings required.
- \+ Intuitive to comprehend for the user.
- \+ No external hardware required (cheap).
- – Can be hard to control the robot.
- – Can be physically tiring for the user.

Kinesthetic Teaching

Video of Kinaesthetic Teaching

Kinesthetic Teaching

Basic Process Flow

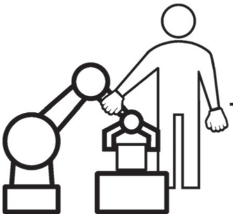


Learning can be performed at multiple levels:

High-level:
- Learning task structure.
- Learning intentionality.
- Symbolic language representation.

Low-Level:
- Learning sensorimotor coupling.
- Learning reusable movement primitives and manipulation skills.
- Typically dynamical system or probabilistic representation.

Aim: Make in-contact tasks easier to program



Task Demonstration → Models of Task → Controller Parameters

Kinesthetic Teaching

The Key to Good Task Execution

Better Demonstrations → Better Learning → Better Execution

Better Demonstrations: The robot should feel *transparent.*

Better Learning: Include task semantics.

Better Execution: Design better controllers.

Consider the parameters of the task.

Kinesthetic Teaching
Demonstration: Problem with Contact Tasks

- How should compliance parameters be chosen?
- Parameter tuning is not easy!

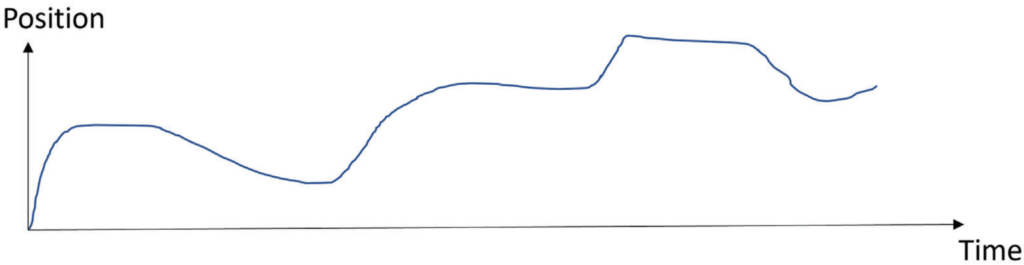
The controller gains need to vary during demonstration dependent on

- Human compliance
- Compliance of environment

There exist many methods for gain scheduling that can be categorized into

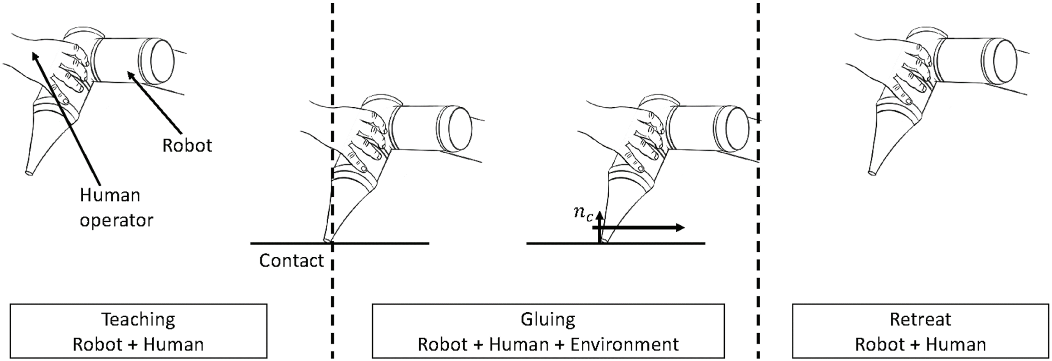
- Velocity-based scheduling
- Force-based scheduling
- Passivity-based scheduling

Kinesthetic Teaching
Segmentation: Learn Semantics of Task



What is going on in this demonstration?

The geometry on the environment can be estimated, e.g. the normal vector to a surface.



Kinesthetic Teaching

Demonstration of Constrained Robot Motion

Summary

A **joint space** dynamic model of an n degree of freedom manipulator is

$$B(q)\ddot{q} + \underbrace{C(q,\dot{q})\dot{q} + g(q)}_{=n(q,\dot{q})} = \tau - J^T(q)h_e$$

An **operational space** dynamic model of an n degree of freedom manipulator is

$$\Lambda(q)\ddot{x} + \Gamma(q,\dot{q})\dot{x} + \eta(q) = h_c - h_e$$

where

$$\begin{aligned} \Lambda(q) &= J^{-T}(q)B(q)J^{-1}(q) \\ \Gamma(q,\dot{q}) &= J^{-T}(q)C(q,\dot{q})J^{-1}(q) - \Lambda(q)\dot{J}(q)J^{-1}(q) \\ \eta(q) &= J^{-T}(q)g(q) \end{aligned}$$

An inverse dynamic control in **joint space** is given by

$$\begin{cases} \tau = B(q)y + C(q,\dot{q})\dot{q} + g(q) \\ y = \ddot{q}_{\mathsf{d}} + K_{\mathsf{D}}(\dot{q}_{\mathsf{d}} - \dot{q}) + K_{\mathsf{P}}(q_{\mathsf{d}} - q), \end{cases}$$

where $K_P, K_D \in \mathbb{R}^{n \times n}$ are positive definite matrices.

An inverse dynamic controller in **operational space** is given by

$$\begin{cases} h_c = \Lambda(q)\alpha + \Gamma(q,\dot{q})\dot{x} + \eta(q) + h_e \\ \alpha = \ddot{x}_{\mathsf{d}} + K_{\mathsf{D}}(\dot{x}_{\mathsf{d}} - \dot{x}) + K_{\mathsf{P}}(x_{\mathsf{d}} - x), \end{cases}$$

where $K_P, K_D \in \mathbb{R}^{n \times n}$ are positive definite matrices.

An impedance control **without contact force measurements** is given by

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) \\ y = J_A^{-1}(q)M_d^{-1}\left(M_d\ddot{x}_d + K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_A(q,\dot{q})\dot{q}\right) \end{cases}$$

An impedance control relying on **contact force measurements** is given by

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) + J^T(q)h_e \\ y = J_{A_d}^{-1}M_d^{-1}\left(K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_{A_d}\dot{q} + M_d\dot{b} - h_e^d\right) \end{cases}$$

An admittance control with orientation given by **Euler angles** is given by

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) + J^T(q)h_e & \text{(Inverse Dynamics)} \\ y = J_A^{-1}M_d^{-1}\left(M_d\ddot{x}_t + K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_A\dot{q}\right) & \text{(Motion Control)} \\ h_e^d = M_t\ddot{\tilde{z}} + K_{Dt}\dot{\tilde{z}} + K_{Pt}\tilde{z} & \text{(Impedance Control)} \end{cases}$$

An admittance control with orientation given by **quaternions** has *Impedance Control* given by

$$\begin{cases} M_p\Delta\ddot{p}_{cd} + D_p\Delta\dot{p}_{cd} + K_p\Delta p_{cd} = f \\ M_o\Delta\dot{\omega}_{cd}^d + D_o\Delta\omega_{cd}^d + K_o'\epsilon_{cd}^d = \mu^d \end{cases}$$

The geometric Jacobian J and analytic Jacobian J_A are related as

$$J = \begin{bmatrix} I & 0 \\ 0 & T(\phi) \end{bmatrix} = T_A(\phi) J_A$$

where (for ZYZ Euler angles)

$$T(\phi) = \begin{bmatrix} 0 & -s_\varphi & c_\varphi s_\vartheta \\ 0 & c_\varphi & s_\varphi s_\vartheta \\ 1 & 0 & c_\vartheta \end{bmatrix}$$

We define the pose error with respect to the desired frame

$$\tilde{x} = -\begin{bmatrix} o_{d,e}^d \\ \phi_{d,e} \end{bmatrix}$$

where $o_{d,e}^d = R_d^T(o_e - o_d)$ and $\phi_{d,e}$ is the vector of Euler angles from rotation matrix R_e^d.
Then

$$\begin{aligned} \dot{\tilde{x}} &= -\begin{bmatrix} R_d^T(\dot{o}_e - \dot{o}_d) - S(\omega_d^d)R_d^T(o_e - o_d) \\ T^{-1}(\phi_{d,e})R_d^T(\omega_e - \omega_d) \end{bmatrix} \\ &= -\underbrace{\begin{bmatrix} I & 0 \\ 0 & T^{-1}(\phi_{d,e}) \end{bmatrix} \begin{bmatrix} R_d^T & 0 \\ 0 & R_d^T \end{bmatrix} J(q)}_{J_{A_d}} \dot{q} + b \end{aligned}$$

where

$$b = \begin{bmatrix} R_d^T \dot{o}_d + S(\omega_d^d) o_{d,e}^d \\ T^{-1}(\phi_{d,e})\omega_d^d \end{bmatrix}$$